ABCB1-Related Acute Transient Encephalopathy (ENPAT): Comprehensive Disease Report

MONDO: MONDO:0975801 · **OMIM:** #620950 · **Gene:** *ABCB1* (P-glycoprotein / MDR1) · **Category:** Genetic (autosomal recessive)

Summary

ABCB1-Related Acute Transient Encephalopathy — catalogued as **"Encephalopathy, acute transient" (ENPAT; OMIM #620950; MONDO:0975801)** — is an ultra-rare, autosomal recessive neurological disorder caused by **biallelic loss-of-function (LoF) variants in *ABCB1***, the gene encoding **P-glycoprotein (P-gp / MDR1)**, the principal ATP-dependent xenobiotic efflux pump of the blood–brain barrier (BBB). When both *ABCB1* alleles are inactivated, the brain loses its active "chemical defense" at the capillary endothelium. The result is a two-hit disease: a constant genetic lesion (P-gp loss) that becomes clinically manifest only when a **trigger** is applied — either a **P-gp–substrate drug** (classically ivermectin) or the **inflammatory milieu of an acute febrile/afebrile illness**. During these triggers, neurotoxic xenobiotics and/or endogenous cytokines accumulate in the CNS and produce a **recurrent but fully reversible encephalopathy** (irritability, visual hallucinations, ataxia, spasticity, pyramidal signs, obtundation, and coma) that resolves within roughly **24–48 hours** with supportive care, leaving normal cognition, imaging, and metabolic studies between episodes.

The entire human evidence base rests on **two independent 2020 reports**: Seo et al. described monozygotic Korean twin sisters with compound-heterozygous LoF *ABCB1* variants and illness-triggered reversible encephalopathy (PMID: 32627353), and Baudou et al. described serious ivermectin toxicity in a patient with *ABCB1* nonsense mutations (PMID: 32813957). This narrow human dataset is powerfully corroborated by decades of cross-species models of the same molecular defect: **mdr1a knockout mice** (100-fold ivermectin hypersensitivity, impaired brain cytokine clearance) and **naturally MDR1-mutant herding dogs** (Collies, Australian Shepherds) that develop ivermectin and loperamide neurotoxicity.

The practical take-home is simple and high-impact: the disorder is **preventable and manageable** through **lifelong strict avoidance of P-glycoprotein-substrate medications** plus prompt supportive care (hydration, antipyresis, symptomatic management) during intercurrent illness. There is no gene- or drug-specific therapy; the "treatment" is risk-avoidance and genetic counseling.

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1. Disease Information

Overview. ENPAT is a monogenic disorder of **CNS xenobiotic clearance**. Affected individuals are developmentally and cognitively normal and are asymptomatic between episodes. Under a trigger (acute illness or a P-gp-substrate drug), they develop acute, self-limited encephalopathy with full recovery. The MedGen/OMIM definition states: *"Acute transient encephalopathy (ENPAT) is characterized by episodes of neurologic dysfunction following treatment with specific medications or onset of acute illness. Neurologic signs include ataxia, spasticity, pyramidal signs, diplopia, vomiting, obtundation, and coma. Symptoms fully resolve in 24 to 48 hours (Baudou et al., 2020; Seo et al., 2020)."*

Key identifiers.

Resource	Identifier
OMIM (phenotype)	#620950 (ENPAT)
MONDO	MONDO:0975801
UMLS	C5975397
MedGen	CUI C5975397 (concept 1874927)
Orphanet	**Not available** (no ORPHAcode on OLS4 cross-reference)
ICD-10/ICD-11	No dedicated code identified
MeSH	No dedicated descriptor identified
Causal gene	*ABCB1* — HGNC:40; NCBI GeneID 5243; Ensembl ENSG00000085563; UniProt P08183; gene OMIM 171050; locus 7q21.12

Synonyms / alternative names. Encephalopathy, acute transient (ENPAT); ABCB1-related / P-glycoprotein-deficiency encephalopathy. *Gene aliases:* MDR1, P-gp/P-170, CD243, GP170, PGY1, ABC20, CLCS, ENPAT.

Information source. The disease-level entity is aggregated from **individual-patient case reports** (two families) into OMIM/MONDO/MedGen. It is therefore a hybrid: ontology-level identifiers built on a very small number of deeply characterized individual patients, supplemented by model-organism data. *(Evidence: F005; PMID: 32627353.)*

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2. Etiology

Primary cause — genetic. The disease is caused by **biallelic (homozygous or compound-heterozygous) loss-of-function variants in *ABCB1***. In the index family, compound-heterozygous LoF variants were identified: a **nonsense allele p.Pro1182X (p.Gln1182Ter)** and a **splice-donor variant c.2786+1G>T** that disrupts exon 23 splicing (producing an 11-bp deletion / exon skipping / intron retention, all frameshifting). Variants segregated from each non-consanguineous parent, were absent from gnomAD and from 1,020 Korean exomes, and left only ~10% residual wild-type protein. *(F001; PMID: 32627353: "revealed compound heterozygous loss-of-function variants in ABCB1".)*

Genetic risk factors. The causal determinant is the biallelic LoF genotype itself. gnomAD contains ~47 heterozygous *ABCB1* LoF alleles but **no homozygotes**, and the estimated carrier-couple probability of two damaging variants is **~2.8×10^{-8}** — explaining the extreme rarity. Heterozygous carriers (including the parents) are clinically unaffected, consistent with recessive inheritance.

Environmental risk factors / triggers. The genetic lesion is necessary but not sufficient — disease requires a **trigger**: 1. **Acute illness** — febrile or afebrile intercurrent infection with fever, vomiting, abdominal pain (the illness-triggered form; Seo et al.). 2. **P-gp-substrate drugs** — classically **ivermectin**; by extension, other P-gp substrates (loperamide, vinblastine, vincristine, digoxin, etc.) are expected hazards (the drug-triggered form; Baudou et al.). *(F002, F006; PMID: 32813957.)*

Protective factors. No genetic protective modifier is defined. The principal **environmental protective factor is avoidance of P-gp-substrate drugs** and prompt management of intercurrent illness. Residual ~10% wild-type P-gp in the index patients may partly explain the reversible, non-fatal course.

Gene–environment interaction. This disease is a textbook **two-hit gene × environment interaction**: a fixed genetic loss of BBB efflux capacity is unmasked only upon exposure to a pharmacologic or inflammatory "second hit." Neither the genotype alone (carriers, interictal patients) nor the trigger alone (in P-gp-competent individuals at standard doses) produces the phenotype.

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3. Phenotypes

Core clinical picture (episodic, illness-/drug-triggered, fully reversible): Both index twins had normal developmental milestones. First acute encephalopathy occurred at age **5 years 6 months**, with recurrent episodes (sometimes simultaneous in both twins), each beginning during acute illness. The stereotyped sequence was: **irritability/agitation → visual hallucinations → progressive drowsiness/obtundation with generalized spasticity and hyperreflexia**, lasting **hours to one day**, with **complete recovery** after hydration and symptomatic care and no seizures or sequelae. At last follow-up (14y3m) growth and cognition were fully normal. *(F004; PMID: 32627353: "recurrent reversible encephalopathy accompanied by acute febrile or afebrile illness".)*

Phenotype	Type	HPO suggestion	Characteristics
Acute/episodic encephalopathy	Clinical sign	HP:0006846 (Acute encephalopathy)	Childhood onset; episodic/ fluctuating; fully reversible in 24–48 h
Irritability / agitation	Behavioral	HP:0000737 / HP:0000713	Early-episode symptom
Visual hallucinations	Symptom	HP:0002367	Mid-episode
Obtundation / coma	Clinical sign	HP:0030151 / HP:0001259	Peak of episode
Spasticity	Clinical sign	HP:0001257	During episode
Hyperreflexia / pyramidal signs	Clinical sign	HP:0001347	During episode
Ataxia	Clinical sign	HP:0001251	Reported (ENPAT definition)
Diplopia	Symptom	HP:0000651	Reported (ENPAT definition)
Vomiting / abdominal pain	Symptom	HP:0002013 / HP:0002027	Accompany triggering illness
Episodic fever	Trigger-associated	HP:0001954	Marks triggering illness

Severity / progression / frequency. Severity is **variable but self-limited** (can reach coma yet reverses fully). Course is **episodic/relapsing with complete interictal recovery** — *not* progressive. Frequency estimates among affected individuals cannot be derived from n=2 families; within affected individuals episodes are recurrent and triggered.

Quality-of-life impact. Between episodes QoL is essentially normal (normal cognition, growth, schooling). Burden derives from unpredictability of episodes, need for vigilant drug avoidance, and risk of severe (coma-level) acute events. No formal EQ-5D/SF-36 data exist.

4. Genetic / Molecular Information

Causal gene. ***ABCB1*** (HGNC:40; GeneID 5243; Ensembl ENSG00000085563; UniProt **P08183** — "ATP-dependent translocase ABCB1 / P-glycoprotein 1," 1280 aa; gene OMIM 171050; 7q21.12). *(F005.)*

Pathogenic variants (from the two human reports and ClinVar).

Variant (HGVS)	Protein	Type	ClinVar	Notes
nonsense (index)	p.Pro1182X / p.Gln1182Ter	Nonsense	3340128	Index twin allele 1
c.2786+1G>T	splice donor, exon 23	Splice-site	3340127	Index twin allele 2; 11-bp del/exon skip/ intron retention, frameshift
NM_001348946.2:c.2380C>T	p.Arg794Ter	Nonsense	3340125	LoF allele
NM_000927.4:c.3053_3056del	frameshift	Frameshift	3340126	LoF allele

Classification. LoF alleles are classified **pathogenic/likely pathogenic** per ACMG/AMP (null variants in a gene where LoF is the disease mechanism; PVS1-type). *(F009.)*

Allele frequency. Index variants were **absent from gnomAD and 1,020 Korean exomes**. gnomAD carries ~47 heterozygous *ABCB1* LoF alleles, **none homozygous**. Carrier-couple probability of two damaging variants ≈ $\mathbf{2.8\times10^{-8}}$.

Origin & functional consequence. Germline, biallelic, **loss-of-function** (not gain-of-function or dominant-negative). Residual ~10% wild-type protein in the index patients. *(F001.)*

Modifier genes / epigenetics / chromosomal abnormalities. None established. No DNA-methylation, histone, or structural/aneuploidy findings are reported for this entity. (Note: common pharmacogenetic *ABCB1* polymorphisms such as C3435T [rs1045642] modulate drug handling and stroke/clopidogrel outcomes [PMID: 32468216; 30742211] but are **not** causes of ENPAT.)

5. Environmental Information

Environmental / pharmacologic factors. The dominant environmental determinant is **exposure to P-glycoprotein-substrate drugs**, above all **ivermectin (CHEBI:6078)** — a classic high-affinity P-gp substrate. In P-gp-deficient individuals, loss of BBB efflux lets ivermectin accumulate centrally and potentiate GABA-/glycine-gated chloride currents, producing CNS depression and coma. *(F006, F008; PMID: 32813957.)* Other P-gp substrates (loperamide, vinca alkaloids, digoxin, many others — *"A number of clinically relevant, structurally and functionally unrelated drugs are substrates for P-gp,"* PMID: 15500562) are presumptive hazards.

Lifestyle factors. No specific diet/smoking/alcohol associations are known. Avoidance of offending drugs is the key behavioral factor.

Infectious agents. No single pathogen causes the disease. Rather, **any acute febrile/afebrile illness** can act as a nonspecific inflammatory trigger; the mechanistic driver is the **cytokine surge**, not a particular organism. *(F002, F004.)*

6. Mechanism / Pathophysiology

Ordered causal chain

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1. Biallelic loss-of-function ABCB1 variants (germline)
        | results in
2. Absent/nonfunctional P-glycoprotein at the luminal membrane of
   brain-capillary endothelial cells (BBB efflux pump lost; ~10% residual WT)
        | leads to
3. Loss of active efflux ("chemical defense") at the blood-brain barrier
        | then -- requires a TRIGGER -- branches:
   +-------------------------------+--------------------------------------+
   | BRANCH A (drug-triggered)     | BRANCH B (illness-triggered)         |
   | 4A. Exposure to P-gp substrate| 4B. Acute febrile/afebrile illness   |
   |     (e.g. ivermectin)         |     -> systemic + CNS cytokine surge |
   |        | results in           |     (IL-1b, TNF, IL-6, CCL2)         |
   | 5A. Drug accumulates in CNS   |        | results in                  |
   |     (verified: brain uptake^) | 5B. Failure to clear cytokines from  |
   |        | leads to             |     brain (shown in Mdr1a/b-/- mice) |
   | 6A. Potentiation of GABA-A /  |        | leads to                    |
   |     glycine Cl- currents ->   | 6B. Sustained neuroinflammatory      |
   |     CNS depression            |     milieu / neuronal dysfunction    |
   +---------------+---------------+------------------+-------------------+
                   | converge on
7. Acute, reversible neuronal dysfunction across cerebral + cerebellar
   cortex and deep gray matter (bilateral)
                   | manifests as
8. Clinical encephalopathy: irritability -> hallucinations -> spasticity,
   hyperreflexia, obtundation, coma
                   | followed by (trigger resolves / drug cleared)
9. Full reversal within 24-48 h, normal interictal state
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Steps 2, 4A, and 5A are **experimentally demonstrated in patients** ([^{11}C]verapamil PET); step 5B is **demonstrated in the knockout-mouse model**; steps 6A/6B/7 are **mechanistically inferred** from pharmacology and animal data.

Supporting detail

Protein dysfunction (upstream lesion). LoF variants eliminate functional P-gp. **P-gp is the major efflux transporter of the BBB** — *"the mdr1a P-glycoprotein is the major P-glycoprotein in the blood-brain barrier and that its absence results in elevated drug levels in many tissues (especially in brain)"* (PMID: 7910522). *(F002.)*

Direct in-patient demonstration of the efflux defect. Dynamic **[^{11}C]verapamil PET** showed markedly increased brain uptake/retention in the twins (target-to-reference ratios **3.27 and 2.81**) versus carrier parents (**1.10** / **0.61**) and the wild-type sister (**0.27**) — *"The patient brains displayed substantial loss of xenobiotic clearance ability, as demonstrated by [^{11}C]verapamil positron emission tomography (PET) study."* Uptake was increased throughout cerebral and cerebellar cortex and deep gray matter (thalamus, basal ganglia), bilaterally, with **normal structural MRI**. *(F002, F007.)*

Cytokine-clearance failure (illness-triggered branch). In **Mdr1a/b$^{-/-}$ mice** given LPS (5 mg/kg IP), brain cytokines (*Il1b, Tnf, Il6, Ccl2*) peaked at 3 h in both genotypes but **failed to clear by 24 h in knockouts** — *"The endogenous cytokine clearance from the brain was also decreased in LPS-treated ABCB1 knockout mice compared to controls."* This provides the mechanistic bridge between an acute illness/cytokine surge and sustained CNS dysfunction. *(F002.)*

Drug-neurotoxicity pharmacology (drug-triggered branch). Ivermectin's invertebrate target is the glutamate-gated chloride channel; in mammals at high brain concentrations it potentiates **GABA-A and glycine receptor chloride currents**, producing CNS depression/coma. A 2026 integrative review concluded that *"dysfunction of P-glycoprotein (ABCB1/MDR1)-mediated efflux [is] a central determinant of ivermectin neurotoxicity,"* with *"severe neurotoxic events, including encephalopathy, seizures, coma, and death, occurring after supratherapeutic exposure and, in susceptible individuals, even at standard therapeutic doses"* (PMID: 41837342). *(F008.)*

Molecular pathways / processes. Core process is **ATP-dependent transmembrane xenobiotic efflux (transport), not classical signaling-cascade dysregulation**. Downstream, neuroinflammatory signaling (IL-1β/TNF/IL-6/CCL2) and inhibitory neurotransmission (GABAergic/glycinergic chloride flux) are implicated.

Suggested ontology terms. GO:0140359 (ABC-type transporter activity); GO:0042493 (response to drug); GO:0006805 (xenobiotic metabolic process); GO:0150104 (transport across blood–brain barrier). **Cell types (CL):** brain microvascular endothelial cell / CL:0002139 (endothelial cell of vascular tree); CL:0000129 (microglial cell, downstream neuroinflammation); CL:0000540 (neuron, final effector). **Cellular component (GO):** GO:0016324 (apical plasma membrane).

7. Anatomical Structures Affected

Organ level. The **brain / central nervous system (UBERON:0000955 brain; UBERON:0001017 CNS)** is the clinically affected organ. No primary non-CNS organ dysfunction is reported in affected humans (metabolic, hepatic, renal labs normal interictally). *(F007.)*

Tissue / cell level. The lesion localizes to the **blood–brain barrier endothelium** — specifically the **luminal (apical) membrane of brain-capillary endothelial cells (brain microvascular endothelial cell)**. P-gp is also physiologically expressed in intestinal epithelium, renal proximal tubule, biliary canalicular membrane, placenta, and adrenal cortex: *"P-gp is expressed by a variety of normal tissues including the intestines, renal tubular cells, brain capillary endothelial cells, biliary canalicular cells, and others, where it functions to actively extrude substrate drugs"* (PMID: 15500562). *(F007.)*

Subcellular level. Apical plasma membrane (GO:0016324) of endothelial cells; P-gp is an integral membrane ATP-dependent transporter (UniProt P08183).

Localization & lateralization. PET showed **bilateral, symmetric** increased tracer retention across cerebral and cerebellar cortex and deep gray nuclei (thalamus **UBERON:0001897**, basal ganglia **UBERON:0002420**, cerebellar cortex **UBERON:0002129**). The process is **diffuse and bilateral**, not focal.

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8. Temporal Development

Onset. Pediatric onset (first reported episode at **5y6m**), consistent with childhood emergence once ambient drug/illness exposures accumulate. Onset of each episode is **acute**, over hours, triggered.

Progression. Non-progressive at the level of the individual: the underlying disorder is static (fixed genetic efflux deficit), and neurological function is normal between episodes. The **disease course is episodic / relapsing with complete remission** between attacks; each episode is **self-limited**, resolving within **24–48 hours**.

Patterns. Remission is **spontaneous / supportive-care-induced** (hydration, antipyresis, symptom management). The key **critical period = the trigger window** (duration of intercurrent illness or presence of the offending drug); intervention during this window (avoidance/removal of trigger, supportive care) is both preventive and therapeutic. *(F004.)*

9. Inheritance and Population

Epidemiology. Ultra-rare — only **two families** reported worldwide (Seo et al. 2020; Baudou et al. 2020). No prevalence or incidence figures exist; no Orphanet epidemiologic class is assigned (no ORPHAcode). Carrier-couple probability of two damaging variants ≈ $\mathbf{2.8\times10^{-8}}$. *(F005, F009.)*

Inheritance. Autosomal recessive (biallelic LoF; unaffected heterozygous parents; variant segregation one allele per parent). **Penetrance is trigger-dependent** — the genotype is fully penetrant for *susceptibility*, but clinical episodes require a trigger (conditional/incomplete clinical penetrance). **Expressivity** appears variable (illness-triggered vs drug-triggered forms; episode severity ranges up to coma). No genetic anticipation (not a repeat-expansion disorder). No founder effect, germline mosaicism, or consanguinity documented (index parents non-consanguineous). Carrier frequency is extremely low (no gnomAD homozygotes). *(F001.)*

Population demographics. Reported patients are **Korean (twin sisters, Seo et al.)** and a patient with ivermectin toxicity (Baudou et al.); no ethnic predisposition can be inferred from n=2. **Sex ratio** indeterminate (index cases female). **Age distribution:** childhood-onset based on available cases.

10. Diagnostics

Clinical / laboratory tests. Interictal metabolic workup is **normal** and serves mainly to exclude mimics: serum amino acids, acylcarnitine profile, urine organic acids, venous blood gas, lactate, creatine kinase — all normal; **CSF without infection/inflammation**; **brain MRI normal**; **EEG nonspecific without electrographic seizures**. *(F004.)*

Imaging / functional biomarker. Dynamic [^{11}C]verapamil PET is the key functional diagnostic, directly demonstrating loss of BBB efflux (increased brain uptake/retention; target-to-reference ratios ~2.8–3.3 in patients vs ~0.3–1.1 in controls/carriers). Structural MRI is normal — a useful discriminator. *(F002.)*

Genetic testing (diagnostic gold standard). Molecular confirmation of biallelic LoF *ABCB1* variants. Recommended approaches: **whole-exome or whole-genome sequencing** (how the index family was solved — "revealed compound heterozygous loss-of-function variants in

ABCB1"), or **targeted *ABCB1* sequencing / gene panels** including copy-number and splice analysis. Chromosomal microarray / karyotype / FISH / mtDNA / repeat-expansion testing are **not applicable** (point/splice LoF mechanism). *(F001, F009.)*

Diagnostic criteria (proposed; no formal guideline exists). (1) Recurrent, fully reversible encephalopathy triggered by acute illness or a P-gp-substrate drug; (2) normal interictal exam, MRI, and metabolic studies; (3) biallelic LoF *ABCB1* variants; (4) supportive functional evidence of reduced BBB efflux (e.g., [^{11}C]verapamil PET).

Differential diagnosis. Metabolic/mitochondrial encephalopathies (excluded by normal metabolic panels/lactate), acute disseminated encephalomyelitis and infectious/autoimmune encephalitis (excluded by normal CSF/MRI), epileptic encephalopathy (no electrographic seizures), hepatic/uremic/toxic encephalopathy, and other drug-toxicity syndromes.

Screening. No newborn screening exists. **Cascade carrier testing** of relatives once a familial variant is identified is appropriate.

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11. Outcome / Prognosis

Survival / mortality. No disease-specific mortality reported in the human cases; episodes were fully reversible. However, mechanism and model data indicate that **severe exposures can cause coma and, by extrapolation from animal/pharmacovigilance data, potentially death** (*"encephalopathy, seizures, coma, and death"*, PMID: 41837342) — so the condition is potentially life-threatening if a high-dose P-gp-substrate exposure occurs. *(F008.)*

Morbidity / function. Between episodes, affected individuals are **functionally normal** (normal growth and cognition at 14y follow-up). No chronic disability was documented. *(F004.)*

Disease course / recovery. Excellent recovery potential — **complete resolution within 24–48 h** with supportive care and trigger removal; no cumulative deficit across recurrent episodes in the index family.

Prognostic factors. Prognosis depends chiefly on **trigger avoidance and prompt recognition**: the main modifiable risk is inadvertent administration of a P-gp-substrate drug. Residual P-gp activity (e.g., ~10% WT in index patients) may favor reversibility.

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12. Treatment

There is **no gene-specific or disease-modifying therapy**. Management is **avoidance + supportive care**.

Supportive / symptomatic care (acute episode). Hydration, antipyresis, and symptomatic treatment (antibiotics for the triggering infection when indicated, antiemetics, antihistamines); episodes resolve without seizures or sequelae. *(F004.)* **NCIT suggestions:** NCIT:C15226 (Supportive Care); NCIT:C70742 (Fluid/Hydration Therapy); NCIT:C25218 (Symptomatic Treatment).

Pharmacotherapy / pharmacogenomics — the central principle. Lifelong strict avoidance of P-glycoprotein-substrate medications, foremost **ivermectin** (and other avermectins, loperamide, vinca alkaloids, digoxin, and numerous other P-gp substrates). This is a **pharmacogenomic contraindication** driven directly by the *ABCB1*-null genotype. *(F006, F008; PMID: 32813957.)*

Advanced / experimental therapeutics. None established. No gene therapy, cell therapy, RNA therapy, targeted therapy, immunotherapy, surgical, or rehabilitative intervention is indicated or reported. No clinical trials (NCT) target this entity.

Treatment strategy (personalized medicine). Genotype-guided prescribing: flag the *ABCB1*-null status in the medical record as an absolute contraindication to P-gp substrates; prophylactically manage intercurrent illness; provide an emergency/anesthesia alert card.

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13. Prevention

Primary prevention. For a known affected individual: **avoid P-gp-substrate drugs** and manage febrile illnesses early and aggressively (hydration, antipyresis) to blunt the cytokine-surge trigger. This is the single most effective, actionable intervention.

Secondary prevention. Early recognition of prodromal symptoms (irritability, hallucinations) during illness/drug exposure enables prompt supportive care and trigger removal, aborting progression to obtundation/coma.

Tertiary prevention. Prevent complications of severe episodes (aspiration, injury during obtundation) with standard supportive/ICU-level care as needed.

Genetic counseling. Autosomal recessive counseling: 25% recurrence risk for carrier couples; **cascade carrier testing** of relatives; discussion of prenatal/preimplantation options once the familial variants are known. Emphasize that heterozygous carriers are unaffected. **No immunization or public-health/environmental intervention** applies beyond drug-exposure avoidance. *(F001, F009.)*

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14. Other Species / Natural Disease

This disease has **strong, naturally occurring and engineered animal counterparts**, which constitute much of the supporting evidence base.

Species	Gene / lesion	Phenotype	PMID
Mouse (*Mus musculus*; NCBI Taxon 10090)	*mdr1a* (Abcb1a) knockout; and Mdr1a/b$^{-/-}$	Viable/fertile; **100-fold ↑ ivermectin sensitivity**, 3-fold ↑ vinblastine; elevated brain drug levels; impaired brain cytokine clearance	7910522
Mouse (CF-1 strain)	Natural P-gp deficiency	Avermectin neurotoxicity at **100-fold lower** doses; deficiency in **intestinal epithelium and brain capillary endothelium**	9144452
Dog (*Canis lupus familiaris*; NCBI Taxon 9615)	*MDR1/ABCB1* **4-bp deletion ΔMDR1 295-298**	Ivermectin & loperamide neurotoxicity; ↑ toxicity of P-gp-substrate chemotherapeutics	15500562; 14765742; 14627096; 11939306

Breeds affected (dog). Herding breeds — **Collies, Australian Shepherds** (and related). In a northwestern-US Collie sample, **35% were homozygous mutant (affected) and 42% carriers** (PMID: 11939306). *"An MDR1 polymorphism in herding breed dogs, including collies and Australian shepherds, has been demonstrated to be the cause of ivermectin sensitivity in these breeds"* (PMID: 15500562). *(F003, F006.)*

Orthologous genes. Mouse *Abcb1a*/*Abcb1b*; canine *ABCB1 (MDR1)* — all orthologs of human *ABCB1*.

Comparative biology. The mechanism is **evolutionarily conserved**: in every species, loss of P-gp removes BBB efflux and produces CNS hypersensitivity to P-gp-substrate neurotoxicants — mechanistically identical to human ENPAT. **Veterinary relevance is high** (ivermectin dosing in herding dogs is a routine clinical concern). **No zoonotic transmission** applies (this is a genetic, non-infectious trait).

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15. Model Organisms

Mouse (mammalian in vivo). The **mdr1a (and Mdr1a/b double) knockout mouse** is the definitive model, directly recapitulating (i) loss of BBB efflux and elevated brain xenobiotic levels, (ii) 100-fold ivermectin hypersensitivity, and (iii) impaired brain cytokine clearance after LPS — modeling both the **drug-triggered** and **illness-triggered** branches. *"Disruption of the mouse mdr1a P-glycoprotein gene leads to a deficiency in the blood-brain barrier and to increased sensitivity to drugs"* (PMID: 7910522). The naturally P-gp-deficient **CF-1 mouse** is a complementary spontaneous model. *(F002, F003.)*

Dog (natural large-animal model). MDR1-mutant Collies / Australian Shepherds provide a **naturally occurring** model of the human disease with a clear drug-trigger phenotype.

Genetic model types available. Knockout (constitutive single and double), and natural loss-of-function alleles (CF-1 mouse, MDR1 dog). Humanized/knock-in or conditional models for ENPAT-specific variants are not reported.

Phenotype recapitulation & limitations. Models faithfully reproduce the **core efflux defect and drug/cytokine accumulation**, strongly supporting causality. Limitations: the mouse "encephalopathy" is read out as drug sensitivity / cytokine kinetics rather than the full human episodic syndrome; the **transient, fully-reversible childhood clinical course** and the specific human triggers are not fully modeled. **Model databases:** MGI (mouse), IMSR/KOMP (strains), OMIA (canine natural disease). *(F003.)*

Mechanistic Model / Interpretation

ENPAT is best understood as a **single-pump "chemical-firewall" failure** at the blood–brain barrier. P-glycoprotein normally sits on the luminal endothelial membrane and pumps lipophilic xenobiotics (and contributes to clearance of inflammatory mediators) back into the

blood. Biallelic *ABCB1* LoF removes this firewall. Because the brain has redundancy and only ~10% residual protein remained in the index patients, the deficit is **clinically silent at baseline** and becomes dangerous only when the system is **loaded** by a second hit:

- **Drug load (Branch A):** a P-gp-substrate drug (ivermectin) that is normally excluded now accumulates centrally and potentiates inhibitory (GABA-A/glycine) chloride currents → CNS depression/coma.
- **Inflammatory load (Branch B):** an acute illness floods the brain with cytokines that, lacking P-gp-assisted clearance, persist (shown directly in knockout mice) → neuroinflammatory encephalopathy.

Both branches converge on **diffuse, bilateral, reversible neuronal dysfunction**, and both resolve once the load is removed — explaining the hallmark 24–48 h reversibility and normal interictal state. This model unifies the two human reports, the PET imaging, the knockout-mouse data, and the canine natural disease into one coherent, cross-species-validated causal chain.

Evidence Base

PMID	Title (abbrev.)	Role	Evidence type
32627353	*Biallelic mutations in ABCB1 display recurrent reversible encephalopathy*	**Defining human report** — genotype, phenotype, [^{11}C]verapamil PET, knockout-mouse cytokine data	Human clinical + model organism
32813957	*Serious Ivermectin Toxicity and Human ABCB1 Nonsense Mutations*	**Second human report** — drug-triggered form	Human clinical
7910522	*Disruption of mouse mdr1a P-gp gene leads to BBB deficiency…*	P-gp is major BBB pump; 100-fold ivermectin sensitivity	Model organism
9144452	CF-1 mice P-gp deficiency	Natural BBB/gut P-gp deficiency → avermectin toxicity	Model organism
15500562	*Therapeutic implications of the MDR-1 gene*	P-gp tissue distribution; canine MDR1 ivermectin sensitivity; many drugs are substrates	Review
11939306	MDR1 allele frequency in Collies	Natural canine disease epidemiology (35% affected, 42% carriers)	Veterinary
14627096 / 14765742	Chemo / loperamide toxicity in MDR1 Collies	Drug-substrate toxicity in natural P-gp deficiency	Veterinary
41837342	Ivermectin neurotoxicity review (2026)	ABCB1/MDR1 efflux failure = central determinant of ivermectin neurotoxicity	Review

Note: Papers on common *ABCB1* polymorphisms in stroke/clopidogrel (e.g., PMID: 32468216, PMID: 30742211) and methamphetamine/ApoE regulation of ABCB1 (PMID: 22426312) are **context / pharmacogenetics**, not evidence for the monogenic ENPAT entity.

Limitations and Knowledge Gaps

1. **Extremely small human dataset (n = 2 families).** Prevalence, incidence, sex ratio, penetrance, expressivity, full phenotypic spectrum, and natural history are poorly constrained. Any frequency statement for individual phenotypes is anecdotal.
2. **No Orphanet code / no ICD / no MeSH descriptor** — the entity is newly minted (2020) and not yet integrated into major epidemiologic/clinical coding systems.
3. **Mechanistic steps 6–8 are inferred**, not directly demonstrated in patients; the exact neurochemical events during a human episode (cytokine vs drug contribution, neurotransmitter effects) have not been measured intra-episode.
4. **Trigger thresholds unknown** — which drugs, at what doses, and which illness severities precipitate episodes in P-gp-null humans is uncharacterized.
5. **Genotype–phenotype relationship** (role of residual ~10% P-gp; whether truly null genotypes are more severe/lethal) is untested.
6. **No therapeutic trials**; management rests entirely on mechanistic reasoning and avoidance.
7. **Model gap:** no mouse/organoid model reproduces the full reversible episodic human syndrome or the specific human variants.

Proposed Follow-up Experiments / Actions

1. **International patient registry & GeneMatcher outreach** to identify additional *ABCB1*-biallelic cases and build genotype–phenotype and natural-history data.
2. **Deep phenotyping during an acute episode** — paired serum/CSF cytokine panels, drug-exposure history, and (where feasible) functional efflux imaging — to confirm the inferred steps (cytokine persistence vs drug accumulation) in humans.
3. **Comprehensive P-gp-substrate "avoid list"** curated from DrugBank/PharmGKB, issued as a patient alert card and EHR allergy/contraindication flag.
4. **iPSC-derived BBB-on-chip / brain-microvascular-endothelial models** carrying the patient variants to test drug/cytokine flux and screen candidate protective strategies.
5. **Humanized or variant knock-in mouse** (ENPAT-specific alleles) with a defined illness/drug challenge to model the reversible episodic phenotype.

6. **Pharmacovigilance mining (FAERS)** for severe ivermectin/loperamide CNS toxicity cases to detect undiagnosed *ABCB1*-null individuals.
7. **ClinVar/gnomAD reanalysis** to refine carrier frequency and search for additional biallelic LoF individuals in large sequencing cohorts.
8. **Formal diagnostic-criteria and genetic-counseling guidance** proposed to ClinGen/ACMG given the clear, actionable pharmacogenomic contraindication.

Evidence-source legend: Human clinical (Seo 2020, Baudou 2020); Model organism (mdr1a/Mdr1a/b-knockout and CF-1 mice); Veterinary natural disease (MDR1 herding dogs); Review/integrative (MDR-1 therapeutics review; 2026 ivermectin neurotoxicity review); Computational (gnomAD/ClinVar/OLS4 queries).
